SMARTHACK XI · PPC "24H TO POWER PPC"

Prosumer Sentinel

One energy agent for Romania. It coordinates home batteries on each low-voltage feeder, watches wind turbines, and commits only what healthy, predictable assets can deliver.

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Midday solar is hitting feeder limits, and nobody can say what each asset should be doing

Network operations

- Rooftop export crosses the limit of an LV feeder at noon.
- The evening peak is imported hours later.
- Home batteries optimise alone, so absorption lands where no headroom is needed.

O&M

- Faults are found after the loss.
- Rules like "wind > 15 km/h and power < 10 kW" fire on every stop, curtailment and maintenance included.
- A broken PV string silently shrinks the flexibility a plan relies on.

Shared root cause: no model of what each asset should do given its weather.

Market size and PPC asset figures are unverified; see Assumptions.

The agent runs by itself from data to action, every 15 minutes

Perceive
Prosumer + wind SCADA, RO weather, OPCOM prices, GIS. Validated on the way in.

→

Estimate
Our models: PV and load P10/50/90, site health, wind normal behaviour, zone demand.

→

Qualify
Each asset gets one status: eligible, derated, excluded or human review.

→

Decide
Per-feeder LP, P10 commitment, O&M top-K per crew type.

→

Act
Battery schedules, commitments, work orders, alerts, park reports, GIS flags.

Every action is traceable

Audit row: model version, input hash, prediction, threshold, status. Realised vs committed is logged on the next tick.

One qualification contract links every model to every decision

A broken PV string lowers a feeder's committed MW. A repair returns capacity to the plan.

LIVE, 3 MINUTES

Demo

Baseline and Sentinel replay the same test week: identical weather, assets, limits and prices. Only the decisions differ.

Production → Demand → Feeders F7 → string failure at site 1042 → O&M park → results.

One model per problem, trained by us, each compared with a simple baseline

Model	Question	Architecture	Baselines	Tier
M1 PVNet	What should this PV produce?	PatchTST-style encoder + own reference window	pvlib, persistence, LightGBM	MUST
M2 LoadNet	What will this home consume?	Same encoder, separate weights	persistence, LightGBM	MUST
M4 SiteHealth	Is this site healthy?	1D-CNN on residuals + 14-day summary	threshold, CUSUM	MUST
M6 WindNBM	Is this turbine normal?	Small GRU + classifier on residuals, park context	WindMon rules	SHOULD
M3 DemandNet	Where will demand peak?	Aggregated M2 scenarios per zone	persistence	SHOULD
M8 GridTwin	Is the GIS feeder/phase right?	Voltage correlation + spectral clustering	GIS as-is	SHOULD

Data: Open-Meteo/ERA5, OPCOM and Energy-Charts for Romania; synthetic prosumers and turbines built on physics. Train 2022–23, validate 2024, test 2025, with 30% of sites and one wind park never seen in training. PVDAQ is non-RO validation. Every number says synthetic RO, real RO or non-RO.

Decisions come from real constraints, and we never sell the nominal number

Per-feeder plan (LP)

$$\Sigma\,(P90 - L10 - \text{charge} + \text{discharge}) + \text{rest} \le G + \text{slack}$$

$$e \ge \text{customer comfort reserve } R$$

$$G = \tau \cdot G_{max} + (1 - \tau) \cdot G_{conservative}$$

Worst plausible export. Low GIS trust only ever tightens the limit. When surplus exceeds headroom, we show the residual slack.

P10 commitment

$$\text{committed} = (1 - h) \cdot Q10(\text{deliverable})$$

$$\text{nominal} \ge \text{available} \ge \text{committed}$$

Conformal-calibrated scenarios replayed through the battery model. h is the telemetry-loss haircut. Realised is measured afterwards, misses published.

O&M triage: capacity is real

$$\text{score} = P(\text{fault}) \times (\text{lost kWh} \times \text{price} + \text{lost flex MWh} \times \text{value}) \times \text{days until next visit}$$

Top-K per day per crew. Sensor faults get a remote check, never a truck. Curtailment gets no work order.

Results on the separate test set

Claim	Baseline	Baseline value	Ours	Label	Run ID
Export peak reduction	self-consumption, tuned per-home rule	pending	pending	synthetic RO	pending
Minutes over limit per feeder-day	same	pending	pending	synthetic RO	pending
P10 hit rate	n/a	n/a	pending	synthetic RO	pending
PV nowcast nMAE, coverage	pvlib, persistence, LightGBM	pending	pending	synthetic RO + PVDAQ	pending
Load forecast nMAE	persistence, LightGBM	pending	pending	synthetic RO	pending
Site faults PR-AUC, hours to detect	threshold, CUSUM	pending	pending	synthetic RO + PVDAQ	pending
Comfort breaches, lei/home/month	default rule	pending	pending	synthetic RO	pending
Wind alerts per turbine-day, recall	WindMon rules	pending	pending	synthetic RO	pending

The simulator is pvlib-based, so a synthetic win is partly circular: PVDAQ sits beside it. Misses are reported as measured.

Where the models are wrong, and what we tried

Questions this slide must answer

- Which fault classes are confused or detected late?
- Which weather and sites have the worst forecast error?
- Where does the P10 commitment miss, and by how much?
- Curtailment vs fault: the confusion matrix, mistakes included.

Findings

pending: team/insights.md and eval runs

Robust by design: it can say "I'm not sure", and it shows why

Hard situations

- **Stale telemetry, sensor fault, data gap:** excluded and removed from the commitment.
- **Wide interval, ensemble disagreement, OOD:** human review, no truck.
- **Wrong GIS:** low trust tightens the limit.
- **A model slips:** its fallback runs, so a panel degrades, never the demo.

Why it decided

- Expected vs actual band for every site.
- Status with its reason on every asset.
- Which sites carried the absorption on a feeder.
- Audit trail from action back to the prediction.

Robustness drills on the test set: gaps, frozen sensors, unit errors. pending: drill results + run ID

What PPC would run, and what we assumed

Value

- PPC pays; prosumers opt in free, with a customer-chosen comfort reserve and override at any time.
- First pilot value: feeder congestion relief.
- Wind: fewer false alarms; we quote lost MWh and lei per event from simulation, not a fleet-wide saving.
- Balancing revenue is not assumed.

Assumptions, not verified

- Battery control is simulated; no hardware was touched.
- Export remuneration β and compensation rules.
- Aggregator rules for small assets; distribution/supply data sharing; GDPR consent.
- Prosumer market figures and PPC wind fleet details.
- Battery cost and cycle life are inputs, not results.

Run it yourself

```
git clone github.com/Andrei9383/smarthack
make reproduce      # data → sim → train → eval → replay
npm ci && npm run build && npm start
```

Built in 24 h

Models, data pipeline, LP, agent and app, with the commit history to show it.

Prior work

Andrei's WindMon (PPC HackITall, April 2026): SCADA schema, quality and anomaly rules as our wind baseline, dashboard layout as a visual reference.

Data

Public or synthetic only, sources cited in the README. Results regenerate from the evaluation script.